Here are the explanations for your Power BI homework on advanced topics in Publishing and Sharing. You can copy this content and save it as a PDF.

Power BI Homework: Publishing and Sharing (Advanced)

How does Power BI handle large datasets in the Online Service, and what is the role of Premium Capacity in this?

Power BI handles large datasets primarily through its in-memory VertiPaq engine, which compresses data for efficient storage and fast querying.[1] In the standard Power BI Service (with a Pro license), there is a **1 GB limit** on the size of a single dataset.[2] This can be a significant constraint for large enterprises.

Power BI Premium Capacity and **Premium Per User (PPU)** licenses address this by providing dedicated resources.[3] They allow for much larger dataset sizes, up to **10 GB** per file upload and up to **400 GB** for a single dataset in some Premium tiers.[4] Premium also enables a "Large semantic model storage format" that can exceed the 10 GB limit, storing data outside of memory in the Power BI Service's data lake, which is loaded on demand.[5] Premium Capacity is essential for organizations with massive data volumes that require a scalable, enterprise-grade solution.[6]

What are the differences between Import mode, DirectQuery, and Live Connection in Power BI Service?

These are the three main data connectivity modes, each with different implications for performance, data freshness, and functionality.

- **Import Mode:**
  - **How it works:** Data is imported into Power BI's in-memory engine.[7]
  - **Pros:** Fast performance for reports and visuals, full Power Query and DAX functionality, and no dependency on the source after refresh.
  - **Cons:** Data is not real-time and must be refreshed manually or on a schedule. There are limits on dataset size (1 GB for Pro).[8]
- **DirectQuery:**
  - **How it works:** Power BI maintains a live connection to the source database and sends queries directly to the source whenever a user interacts with a visual.[9]
  - **Pros:** Data is always current, reports are not limited by dataset size, and security is handled at the source.

    - **Cons:** Slower performance for reports due to round trips to the source, and some Power Query and DAX functionality is limited. It requires a stable, high-performance data source.
- **Live Connection:**
    - **How it works:** Similar to DirectQuery, but it connects specifically to a pre-existing Analysis Services model (either on-premise or in Azure).[10]
    - **Pros:** Leverages the power of the Analysis Services engine for fast query performance, inherits the security model and calculations from the source, and reports are always live.
    - **Cons:** No data modeling is possible in Power BI Desktop; all modeling must be done in the source model. It's only for specific data sources.

Explain deployment pipelines in Power BI Online. What stages do they include?

Deployment pipelines are a feature in Power BI Premium and PPU that provide a structured workflow for content developers to manage the lifecycle of their Power BI assets.[11] They help streamline the development, testing, and production process, reducing manual errors and improving collaboration.[12]

The three standard stages are:

1. **Development:** This is the starting point where content creators build and modify reports, dashboards, and datasets.[13] It's the sandbox for new features and changes.
2. **Test:** In this stage, the content is deployed to a controlled environment for testing.[14] Testers and stakeholders can validate the data, functionality, and user experience without impacting the live production environment.
3. **Production:** The final stage where the tested and validated content is made available to end-users.[15] This is the live environment that business users interact with.

How can Power BI Service integrate with Microsoft Teams or SharePoint for collaboration?

- **Microsoft Teams:**
    - You can add a Power BI tab to a Teams channel, which allows you to embed a live, interactive report.[16]
    - Users can then view, discuss, and collaborate on the data directly within Teams.[17]

    - Power BI also has a native app for Teams that provides a full experience, allowing users to browse their reports, workspaces, and apps without leaving Teams.[18]
- **SharePoint:**
    - The Power BI Service offers a SharePoint Online web part that can be used to embed reports directly into a SharePoint page.[19]
    - This allows users to view and interact with live Power BI content within their organization's intranet or document management system, combining data insights with other relevant content.

What is the XMLA endpoint in Premium and how does it benefit developers or enterprise BI teams?

The **XMLA (XML for Analysis) endpoint** is a feature in Power BI Premium and PPU that allows for open-platform connectivity to the semantic model.[20] It exposes the underlying Analysis Services tabular model, enabling external tools to connect to it.[21]

Benefits for developers:

- **Advanced Authoring:** Developers can use external tools like SQL Server Management Studio (SSMS) or DAX Studio to query, manage, and optimize semantic models.[22]
- **Automated Deployment:** It allows for programmatic deployment of models and automation of tasks like scheduled refreshes using PowerShell scripts or other tools.[23]
- **Version Control:** Teams can use source control systems like Git to manage their Power BI projects, which is crucial for enterprise-level development.[24]
- **Backup and Restore:** It enables the backup and restore of semantic models, which is essential for disaster recovery and data management.

Describe how usage metrics and audit logs work in Power BI Service.

- **Usage Metrics:** These are pre-built reports that provide insights into how your reports and dashboards are being used.[25] You can view them to understand:
    - How many views a report has.
    - Who the most active users are.
    - Which pages are most popular.
    - They are useful for identifying popular content, removing unused reports, and proving the value of your BI assets.[26]

- **Audit Logs:** These are detailed records of user activities in the Power BI Service, accessible through the Microsoft 365 Security & Compliance Center.[27] They provide a comprehensive audit trail of actions like:
    - Report and dashboard creation, deletion, or modification.
    - Sharing events.
    - Changes to security settings.
    - Audit logs are crucial for security, compliance, and governance, as they allow administrators to monitor for suspicious activity and maintain a record of all user actions.[28]

How do you manage workspace access and permissions for different users?

Workspace access is managed by assigning **roles** to users or security groups.[29] Each role has a different level of permission:

- **Admin:** Full control over the workspace, including adding/removing users, deleting content, and managing permissions.[30]
- **Member:** Can create, edit, and delete content, publish apps, and manage permissions for other users (except Admins).
- **Contributor:** Can create, edit, and delete content in the workspace but cannot manage permissions or publish apps.
- **Viewer:** Can only view and interact with the content in the workspace.[31] This is the consumer-level role.

You can manage these permissions by going to the workspace settings in the Power BI Service.[32]

How can data governance be enforced in Power BI Service?

Data governance ensures that data is consistent, accurate, and secure.[33] Power BI Service provides several tools to enforce it:

- **Certification and Promotion:** Data owners can **certify** semantic models and dataflows to mark them as "authoritative" or "official" for the organization.[34] This helps users find trusted data sources. Less formal content can be "promoted" to indicate that it's worth using.
- **Centralized Dataflows:** Centralizing data preparation in a dataflow ensures that all reports use the same clean, transformed data.

- **Access Control:** Using workspace roles, apps, and Row-Level Security ensures that only authorized users can access sensitive data.[35]
- **Audit Logs:** Administrators can monitor user activities to ensure compliance with data policies and detect unauthorized sharing or data misuse.[36]
- **Administrative Policies:** Tenant administrators can set global policies for sharing, publishing, and other features to maintain control over the entire environment.

What are the limitations of Row-Level Security when using DirectQuery or Live Connection?

- **DirectQuery:** RLS is supported, but it's important to know that the DAX filter expression is translated into a native query (e.g., SQL) and pushed to the source database. This requires careful consideration of the source database's performance. The DAX `username()` and `userprincipalname()` functions are used to pass the current user's identity to the query.[37]
- **Live Connection:** RLS is **not** defined in Power BI Desktop. Instead, it must be **defined and managed in the source Analysis Services model**.[38] Power BI simply inherits the security roles from the source. The Power BI Service does not show a "Security" option for live connection datasets.[39] This is a key limitation, as it requires you to manage security in two different systems.

Explain how you can refresh a dataset via Power Automate or REST API.

- **Power Automate:** You can create a flow with a trigger (e.g., a file is updated in SharePoint, an email arrives, or on a specific schedule) that includes the **"Power BI - Refresh a dataset"** action.[40] This action connects to the Power BI Service and triggers a refresh of a specified dataset. It is ideal for event-driven refreshes.
- **REST API:** For more technical, automated solutions, you can use the Power BI REST API to programmatically refresh a dataset.[41] You send an `HTTP POST` request to a specific endpoint (e.g., `https://api.powerbi.com/v1.0/myorg/groups/{group_id}/datasets/{dataset_id}/refreshes`). This method is commonly used by developers to integrate dataset refreshes into their ETL processes, scripts, or continuous integration/continuous deployment (CI/CD) pipelines.[42]